Alpha decay 1

Let $V(r)$ be the Coulomb potential for alpha decay where r_1 is the radius of the parent nucleus. The factor $2Z$ is for the two protons of the alpha particle and the Z protons of the daughter nucleus.

$$V(r) = \begin{cases} -V_0 & 0 < r < r_1 \\ \dfrac{2Z\alpha\hbar c}{r} & r > r_1 \end{cases}$$

For an alpha particle with energy E let r_2 be the radius such that

$$V(r_2) = \frac{2Z\alpha\hbar c}{r_2} = E$$

Hence the region from r_1 to r_2 is a barrier potential for an alpha particle with energy E.

Using the WKB method we have for tunneling probability T

$$T \approx e^{-2\gamma}$$

where

$$\gamma = \frac{\sqrt{2m}}{\hbar} \int_{r_1}^{r_2} \sqrt{V(r) - E}\, dr$$

Factor out $\sqrt{E}$ to obtain

$$\begin{aligned}\gamma &= \frac{\sqrt{2mE}}{\hbar} \int_{r_1}^{r_2} \sqrt{\frac{V(r)}{E} - 1}\, dr \\ &= \frac{\sqrt{2mE}}{\hbar} \int_{r_1}^{r_2} \sqrt{\frac{r_2}{r} - 1}\, dr\end{aligned}$$

Apply change of variable from r to u where

$$\begin{aligned} r &= r_2 \sin^2 u \\ dr &= 2r_2 \sin u \cos u \\ u(r) &= \arcsin \sqrt{r/r_2} \\ u_1 &= u(r_1) = \arcsin \sqrt{r_1/r_2} \\ u_2 &= u(r_2) = \arcsin 1 = \frac{\pi}{2} \end{aligned}$$

Hence

$$\begin{aligned}\gamma &= \frac{\sqrt{2mE}}{\hbar}\, 2r_2 \int_{u_1}^{u_2} \sqrt{\frac{1}{\sin^2 u} - 1}\, \sin u \cos u\, du \\ &= \frac{\sqrt{2mE}}{\hbar}\, 2r_2 \int_{u_1}^{u_2} \sqrt{1 - \sin^2 u}\, \cos u\, du \\ &= \frac{\sqrt{2mE}}{\hbar}\, 2r_2 \int_{u_1}^{u_2} \cos^2 u\, du\end{aligned}$$

Solve the integral.

$$\gamma = \frac{\sqrt{2mE}}{\hbar}\, 2r_2 \left(\frac{u}{2} + \frac{\sin(2u)}{4} \right) \Bigg|_{u=u_1}^{u=u_2} \tag{1}$$

Evaluate the limits.

$$\gamma = \frac{\sqrt{2mE}}{\hbar}\, 2r_2 \left(\frac{\pi}{4} - \frac{1}{2} \arcsin \sqrt{r_1/r_2} - \frac{1}{4} \sin\left(2 \arcsin \sqrt{r_1/r_2} \right) \right) \tag{2}$$

From the identities

$$\sin(2x) = 2 \sin x \cos x$$

and

$$\cos(\arcsin x) = \sqrt{1 - x^2}$$

the last term in (2) simplifies as

$$\frac{1}{4} \sin\left(2 \arcsin \sqrt{r_1/r_2} \right) = \frac{1}{2} \sqrt{r_1/r_2} \sqrt{1 - r_1/r_2}$$

Hence

$$\gamma = \frac{\sqrt{2mE}}{\hbar}\, 2r_2 \left(\frac{\pi}{4} - \frac{1}{2} \arcsin \sqrt{r_1/r_2} - \frac{1}{2} \sqrt{r_1/r_2} \sqrt{1 - r_1/r_2} \right)$$

By equation (4) below we have $r_1 \ll r_2$ hence we introduce the approximations

$$\arcsin \sqrt{r_1/r_2} \approx \sqrt{r_1/r_2}$$

and

$$\sqrt{r_1/r_2} \sqrt{1 - r_1/r_2} \approx \sqrt{r_1/r_2}$$

Then γ simplifies as

$$\gamma = \frac{\sqrt{2mE}}{\hbar}\, 2r_2 \left(\frac{\pi}{4} - \sqrt{r_1/r_2} \right)$$

Distribute r_2.

$$\gamma = \frac{2\sqrt{2mE}}{\hbar} \left(\frac{\pi r_2}{4} - \sqrt{r_1 r_2} \right)$$

Substitute

$$r_2 = \frac{2Z\alpha\hbar c}{E}$$

to obtain

$$\gamma = C_1 \frac{Z}{\sqrt{E}} - C_2 \sqrt{Z r_1} \tag{3}$$

where

$$C_1 = \sqrt{2m}\pi\alpha c = 7.9 \times 10^{-7}\ \mathrm{J}^{1/2}$$

and

$$C_2 = 4\sqrt{\frac{m\alpha c}{\hbar}} = 4.7 \times 10^{7}\ \mathrm{m}^{-1/2}$$

For the alpha decay process ${}^{238}_{92}\mathrm{U} \to {}^{234}_{90}\mathrm{Th}$ we have

$$\begin{aligned} A &= 238 \\ Z &= 90 \\ E &= 4.27\,\mathrm{MeV} \end{aligned}$$

Hence

$$\begin{aligned} r_1 &= A^{1/3} \times 1.2 \times 10^{-15}\,\mathrm{m} && = 7.44 \times 10^{-15}\,\mathrm{m} \\ r_2 &= \frac{2Z\alpha\hbar c}{E} && = 60.7 \times 10^{-15}\,\mathrm{m} \end{aligned} \tag{4}$$

Then for γ we obtain

$$\gamma = 47.8$$

For tunneling probability T we have

$$T = e^{-2\gamma} = 3.17 \times 10^{-42}$$

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